

RAMIZA MALIHA

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SUMMARY

MSc student at Uppsala University with a background in backend engineering and research in algorithmic graph mining. Proficient in Java, C++, and Python with a deep understanding of transformer and Diffusion architectures. I have worked on scalable software systems and co-authored a paper for KDD 2024.

EDUCATION

Uppsala University, Uppsala, Sweden.
MSc in Computer Science

Sep 2024 - Present

North South University, Dhaka, Bangladesh.
BS in Computer Science and Engineering

Sep 2014 - Dec 2019
CGPA-3.64/ 4.00 (Graduated Cum Laude)

RELEVANT COURSEWORK

Machine Learning, Deep Learning, Data Engineering, Large Language Models, Statistical Machine Learning, Introduction to Image Analysis, Data Mining, Software Engineering and Project Management, Mathematical Modelling of Football.

EXPERIENCE

Uppsala University, Uppsala, Sweden
Teaching Assistant

Aug, 2024 - Present

- Assisted in labs and project supervision and provided technical guidance to students.
- Subjects: Programming Language (Python, C++), Data Structure, Cryptology

BJIT Limited, Dhaka, Bangladesh
Software Engineer

May, 2022 - July, 2024

- Developed and integrated 10+ features for the Enovia PLM system to support a Finnish paper technology client.
- Built REST APIs using Spring Boot and automated deployments with Jenkins, reducing release cycles by 50%.

North South University, Dhaka, Bangladesh
Research Assistant (RA)

January 2020 -August 2024

Supervisor: [Dr. Ahsanur Rahman](#)

- Co-developed an exact algorithm for the NP-complete dense subgraph enumeration problem using pruning optimizations, achieving speedups over ODES and PCE of 38.6x/6.5x on real graphs and 39.7x/3.1x on synthetic graphs.
- Optimized vertex cover algorithms by researching and implementing approximation techniques that reduced computational runtime. Implemented and benchmarked algorithms in C, C++, and Python, using efficient data structures and mathematical insights from combinatorics and complexity theory.
- Validated algorithmic performance by conducting extensive benchmarking and statistical analysis on both real-world and synthetic datasets.

SELECTED PROJECTS

Generative Probabilistic Frameworks for Constrained Structural Inference in Single-Cell Lineages

- Designed a GFlowNet-based framework that samples diverse, biologically valid lineage trees proportional to a gene expression reward signal
- Implemented a neuro-symbolic constraint layer enforcing acyclicity and single-parent rules directly in the action space, achieving 100% structural validity across all experiments
- Trained on synthetic data and evaluated on four real single-cell datasets (LARRY haematopoiesis, pancreatic endocrinogenesis, Bowling 2020) in a zero-shot transfer setting
- Multi-dataset curriculum training improved biological ordering scores to above 0.99 on unseen tissue types without fine-tuning

JobCraft AI — LLM-Powered Job Matcher [🔗](#)

- Investigated retrieval-augmented generation (RAG) and semantic search for document understanding.
- Implemented a prototype system for job–CV matching and evaluated retrieval quality using cosine similarity metrics.
- Proposed and implemented a rule-based task selection module inspired by agentic reasoning frameworks to improve interpretability and autonomy in LLM-driven review tasks.

Automated Hydroponic System Using IoT and Cloud Computing

- Led a team of three to build an automated hydroponic system that can be monitored remotely.
- Integrated NodeMCU, Arduino, and various sensors connected to the ThingSpeak platform to track real-time data.

PolicyOverPills — Global Health Insights through Tobacco Policy Analysis

- Analysed WHO datasets (2008–2018) covering smoking trends, tobacco policies, GDP, and life expectancy across 120+ countries.
- Engineered behavioral and policy features such as PCA-based tobacco behavior, and applied K-Means clustering to group countries by health outcomes.
- Evaluated decade-long policy effectiveness using ANOVA and cluster shift analysis, identifying that policy-driven smoking reduction correlates with higher life expectancy.

A Star Predictor app using Webis-TLDR-17 Corpus dataset

- Built a scalable data pipeline using Python, Docker, and RayTune to analyze 3.8M+ posts from the Webis-TLDR-17 corpus for trend detection.
- Automated CI/CD pipeline across three servers using Ansible and integrated RabbitMQ for message queuing, reducing deployment time by 40%.

TrendScape—Reddit Discourse Analysis Pipeline

- Built a distributed text analytics pipeline using Spark MLlib for discourse analysis.
- Implemented topic modeling and sentiment scoring modules, experience applicable to evaluating tone and consistency in large text datasets.

Alex the Bartender

- Multi-modal LLM pipeline combining vision and language models to create an empathetic virtual agent.
- Synced visual emotion cues with textual input to generate context-aware responses using Gemini API and Furhat SDK.

Predicting Chronic Kidney Disease (CKD) using Machine Learning

- Leveraged Random Forest and SVM algorithms on real-world clinical data to classify Chronic Kidney Disease stages.
- Executed feature engineering and interpretability analysis to pinpoint biomarkers affecting disease progression.
- Ensured model transparency and adhered to ethical data standards essential for medical AI deployment.

TECHNICAL SKILLS

AI / ML	Generative AI, NLP, Graph Mining, Data Mining, PyTorch, Scikit-learn, TensorFlow
Data / Visualization	R, Matlab, Pandas, NumPy, Matplotlib, NetworkX, Cytoscape, NetworkKit
Big Data Systems	Apache Spark, Hadoop, HDFS
Cloud / DevOps	Docker, Jenkins, Ansible, RabbitMQ, Git, Jira, Trello
Frameworks	Spring Boot, Flask, FastAPI, REST APIs
Databases	MySQL, MongoDB
HCI	Multimodal interfaces, empathetic agents, Furhat Robotics
Tools	VS Code, GNU Parallel, GProf, LaTeX, Google Workspace

PUBLICATION

A Fast Exact Algorithm to Enumerate Maximal Pseudo-cliques in Large Sparse Graphs

DOI: [10.1145/3637528.3672066](https://doi.org/10.1145/3637528.3672066)

- Generated datasets using NetworkX and NetworkKit to test the algorithm's limits.
- Ran benchmarks to validate speedup over existing methods on real-world and synthetic graphs.
- Analyzed graph properties with Cytoscape and wrote the experimental results and related sections of the paper